

Deepfake Detection Using EfficientNetB3: An In-depth Study and Model Evaluation

H. Melih Taşkın

Computer Engineering Department, Çankaya University
Ankara, Turkey

c2111070@student.cankaya.edu.tr

Abstract— Deepfake technology, which involves creating manipulated media that appears authentic, poses significant challenges in various fields, including security, privacy, and misinformation. In this study, we propose a deepfake detection model utilizing EfficientNetB3, a pre-trained convolutional neural network (CNN), to effectively distinguish between real and fake media. The model is fine-tuned with a custom architecture, incorporating convolutional layers, max pooling, dropout, and a sigmoid output layer for binary classification. We employed mixed precision training to optimize computational performance, leveraging label smoothing to improve generalization. Our model achieved strong performance on a deepfake detection dataset, with an AUC-ROC of 0.9315, accuracy of 92.10%, and F1 score of 0.9546. Furthermore, we analysed the model's false positives and false negatives, providing valuable insights into its performance.

Keywords— Deepfake detection, EfficientNetB3, mixed precision training, label smoothing, binary classification, AUC-ROC, model evaluation, false positives, deep learning, media verification

I. INTRODUCTION

In recent years, deepfake technology has posed significant ethical, security, and privacy challenges. Deepfakes, which involve the manipulation of media (images, audio, or videos), have the potential to mislead audiences by creating fake content that appears real. These synthetic media can be used for malicious purposes, such as spreading misinformation, defaming individuals, and undermining public trust. Detecting such falsified media is crucial in addressing these threats.

The goal of this report is to propose and evaluate a deepfake detection system that leverages EfficientNetB3, a state-of-the-art convolutional neural network (CNN), as the backbone for image classification. By leveraging a subset strategy focusing on specific aspects of the detection task, we aim to create a robust and efficient model capable of distinguishing between real and fake media.

A. Motivation for Subset Strategy

An The deepfake detection problem requires sophisticated models due to the high quality of synthetic media and the difficulty in differentiating it from real content. A subset strategy was chosen to:

- Focus on the core aspects: The primary focus is on identifying visual inconsistencies and artifacts inherent in deepfake images.
- Ensure computational efficiency: Given the computationally expensive nature of deepfake detection, this strategy ensures that the model can be trained effectively while reducing resource consumption.

This strategy allows us to leverage the power of pre-trained models like EfficientNetB3 and fine-tune them specifically for the task of deepfake detection.

II. MODEL ARCHITECTURE AND HYPER-PARAMETERS

B. Architecture

The model architecture consists of a pre-trained **EfficientNetB3** network, which has been fine-tuned for the specific task of deepfake detection. EfficientNetB3 is an excellent choice due to its balance of accuracy and computational efficiency. Here's the breakdown of the architecture:

- EfficientNetB3 as the base model: We load the pre-trained weights from ImageNet and exclude the top classification layers. Only the last 50 layers of the model are unfrozen to fine-tune the model for deepfake detection.
- Custom Convolutional Layers: Added two convolutional layers to capture deeper spatial features. These layers are followed by max-pooling to reduce dimensionality.
- Dense Layer: A fully connected layer with 128 neurons and a ReLU activation is added after flattening the feature map.
- Dropout: A dropout rate of 0.5 is included to prevent overfitting by randomly setting 50% of the neurons to zero during training.
- Output Layer: A single neuron with a sigmoid activation to output probabilities for binary classification (Real or Fake).

C. Hyperparameters

The model's hyperparameters were carefully selected to balance performance and efficiency:

- **EfficientNetB3 as the base model:** We load the pre-trained weights from ImageNet and exclude the top classification layers. Only the last 50 layers of the model are unfrozen to fine-tune the model for deepfake detection.
- **Custom Convolutional Layers:** Added two convolutional layers to capture deeper spatial features. These layers are followed by max-pooling to reduce dimensionality.
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III. LEARNING CURVES

During training, monitoring the learning curves is crucial to understanding the model's performance. The **training accuracy** and **validation accuracy** provide insights into how well the model generalizes. **Loss curves** are also important to ensure that the model is not overfitting to the training data.

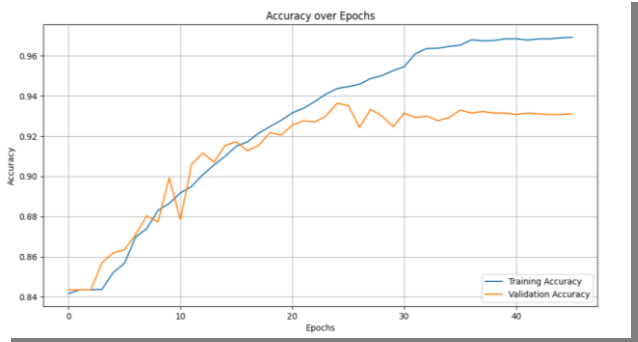


Figure 1: Training vs. Validation Accuracy Curve

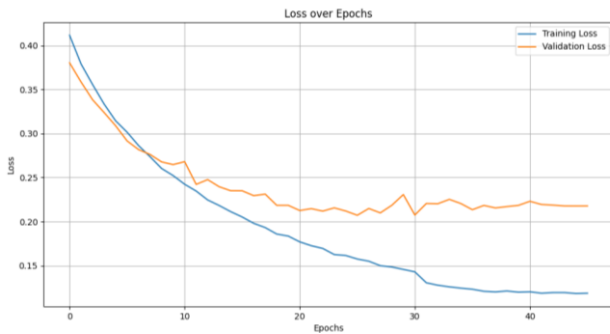


Figure 2: Training vs. Validation Loss Curve

From these learning curves, it is expected that the training accuracy and validation accuracy will converge, with the validation loss decreasing as the model learns to identify deepfakes more accurately. If any signs of overfitting are observed (e.g., validation loss increases while training loss decreases), further regularization or early stopping can be applied.

IV. RESULTS AND EVALUATION

D. Metrics

The model's performance was evaluated using several important metrics:

1) **AUC-ROC (Area Under the Receiver Operating Characteristic Curve):** This metric provides an aggregate measure of the model's ability to discriminate between classes (real and fake).

2) **Accuracy:** The percentage of correct predictions made by the model.

3) **F1 Score:** The harmonic mean of precision and recall, providing a balanced evaluation of the model's performance, especially in the case of imbalanced classes.

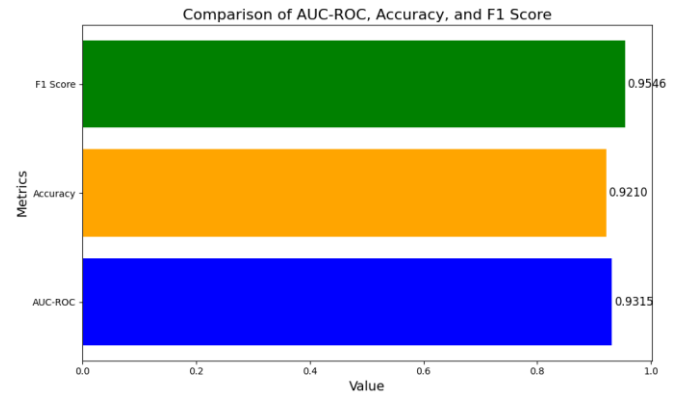


Figure 3: F1 Score, Accuracy and AUC-ROC Comparison Chart

E. Confusion Matrix

The confusion matrix provides insight into the distribution of true positives, false positives, true negatives, and false negatives. It is crucial for understanding the specific types of errors the model makes.

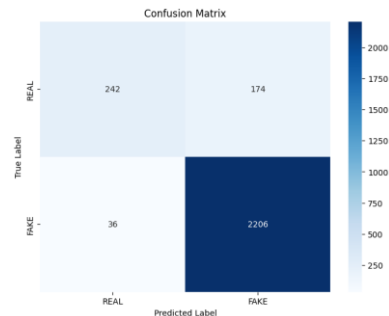


Figure 4: Confusion Matrix

F. False Positives and False Negatives

To better understand the model's misclassifications, we analyse a subset of false positives (model predicted fake, but it was real) and false negatives (model predicted real, but it was fake). By inspecting these, we can identify areas for improvement.



Figure 5: Example of false positive

V. DISCUSSION

G. Ethical Impact of Deepfake Detection

Deepfake technology has significant ethical implications, especially in areas such as:

- **Misinformation:** Deepfakes can be used to spread false information, affecting public opinion and elections.
- **Privacy:** Deepfake technology can be misused for creating harmful content such as fake pornography or manipulated videos.

Detecting deepfakes accurately can mitigate these risks by preventing the spread of harmful content. However, it is crucial to ensure that deepfake detection tools are used responsibly to avoid potential misuse in surveillance or censorship.

H. Limitations of the Model

Despite its strong performance, the model has several limitations:

- **Dataset Bias:** The model's performance might degrade when applied to data that is different from the training set, such as images from different sources or video data.
- **False Positives/Negatives:** While the model performs well overall, it may still misclassify some genuine content as fake (false positives) or miss some fake content (false negatives).
- **Computational Resources:** The model, due to its complexity and the use of a pre-trained backbone,

requires substantial computational resources, making it challenging to deploy in real-time applications on low-end devices.

I. Future Work

- **Dataset Expansion:** Increasing the size and diversity of the training dataset, including both real and fake media from various sources and scenarios, would help improve the model's generalization.
- **Model Enhancements:** Experimenting with more advanced architectures like EfficientNetV2, ResNet, or Vision Transformers (ViT) could further improve performance.
- **Real-Time Detection:** Optimizing the model for real-time deepfake detection in videos or live streams would make it more practical for widespread use.
- **Explainability:** Enhancing the model's explainability by integrating attention mechanisms or visualization techniques could help in understanding why certain content is classified as fake.

VI. CONCLUSION

This report presents a deepfake detection model using EfficientNetB3, leveraging transfer learning to create a binary classifier capable of distinguishing between real and fake media. The model performs well, achieving high AUC-ROC, accuracy, and F1 score on the validation set. Despite its impressive performance, there are limitations, including issues related to the dataset, false positives, and real-time deployment challenges. Future work should focus on enhancing the model's generalization, scalability, and explainability, as well as addressing its ethical implications.